A Novel Approach to Estimating the Real-Time Interbank Rate of USD/TND in the Tunisian FX Market

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Authors: Ahmed Aziz Jaidane | Hadil Heni | Yasmine Fakhfakh

Professor: Dr. Eymen Errais

Abstract

This report develops a hybrid forecasting framework that estimates the intrinsic USD/TND interbank rate by integrating macroeconomic logic with machine learning and stochastic modeling. The model combines a currency basket valuation method with a dynamic regime-switching system leveraging an LSTM neural network and an Ornstein-Uhlenbeck jump diffusion process. The result is a decision engine that adapts to monthly volatility regimes and enforces pip-based consistency checks to yield stable, economically interpretable predictions.

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1. Introduction

Forecasting exchange rates is a long-standing challenge in international finance. In Tunisia, the USD/TND rate is particularly volatile due to seasonal imbalances, liquidity constraints, and regulatory dynamics. The goal of this project is to compute the real-time intrinsic interbank rate using both global FX trends and local spread behavior.

The model is composed of three parts:

1. *A weight basket estimator and a volatility-aware spread forecasting model.*
2. *A Long Short-Term Memory Non-Parametric model.*
3. *A hybrid decision engine.*

This methodology enables traders and analysts to determine the real-time relative value (interbank rate) of the USD/TND in the Tunisian local FX market, providing a benchmark to assess pricing fairness at any point in time.

2. Data Collection & Preparation

Data was consolidated from multiple exchange rate datasets to form a comprehensive series including:

- *Latest Fixing*
- *Spot USD/TND*
- *Daily percentage changes for EUR/USD, GBP/USD, USD/JPY*

A cleaned and standardized dataset is stored in Data.xlsx, containing all variables used across the pipeline. Observations cover a multi-year period (2020–2025), with full temporal alignment.

3. Basket-Based Valuation Model

3.1 Theoretical Equation

This formulation assumes a weighted linear exposure to global currencies. Weights are not assumed arbitrarily but estimated from data.

The model computes:

$$USD/TND\_intrinsic = Latest\ Fixing \times (w1 \times \Delta EUR/USD + w2 \times \Delta GBP/USD + w3 \times \Delta USD/JPY)$$

3.2 Exploratory Analysis

Initial analysis confirmed:

- *Negative relationship between USD/TND and EUR/USD, GBP/USD*
- *Positive correlation with USD/JPY*

Correlation matrices and scatterplots suggested non-linearity, rejecting simple OLS regression.

	Latest Fixing	*EUR/USD*	*GBP/USD*	USD/JPY	*Spot USD/TND*
Latest Fixing	1				
EUR/USD	-0.904670273	1			
GBP/USD	-0.73454131	0.890935	1		
USD/JPY	0.908259604	-0.77491	-0.59051	1	
Spot USD/TND	0.993113086	-0.91677	-0.75826	0.897545	1

Fig 1: Correlation Matix (Currency pairs)

	Var EUR/USD	Var GBP/USD	Var USD/JPY	Spot/Fixing
Var EUR/USD	1			
Var GBP/USD	0.714926755	1		
Var USD/JPY	-0.423189404	-0.187572055	1	
Spot/Fixing	0.021284481	0.016506456	0.004260565	1

Fig 2: Correlation Matix (Daily Variations of Currency pairs)

3.3 SHAP + XGBoost Weight Discovery

Using an XGBoost model, we fitted Spot USD/TND against the FX deltas. SHAP values, in absolute terms, were extracted to measure feature contributions:

- *ΔEUR/USD → 15%*
- *ΔGBP/USD → 12%*
- *ΔUSD/JPY → 13%*

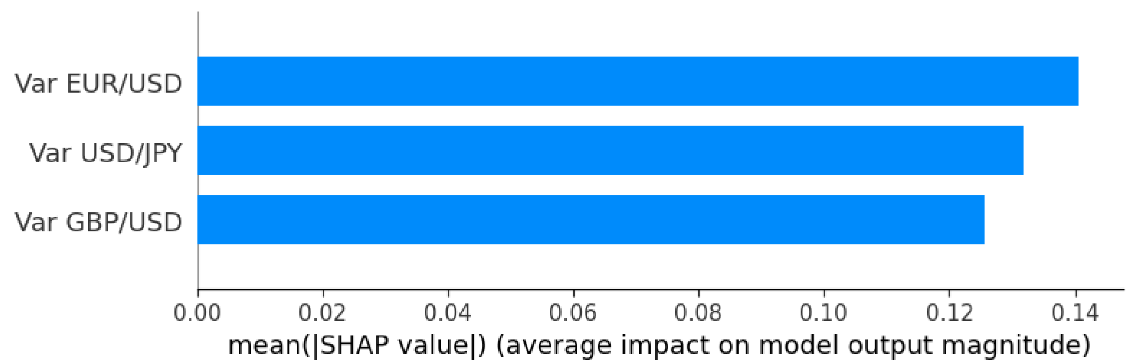

Fig 3: Features importance (XGBoost, Original data)

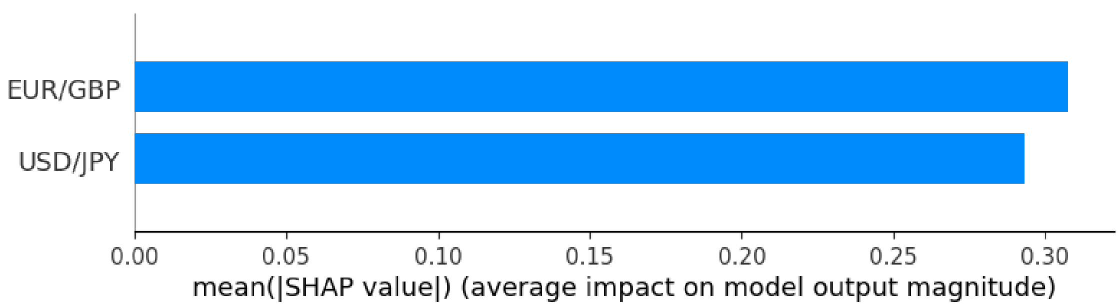

Fig 4: Features importance (XGBoost, Feature-Engineered data)

3.4 Macroeconomic & PCA Alternatives

A macroeconomic approach using FX reserve compositions was considered. PCA was also tested, with explained variance ~64%, 28%, 8% across principal components, but lacked direct interpretability.

4. Spread Modeling & Forecasting (Part II)

4.1 Spread Construction

Spread is defined as:

A monthly breakdown revealed seasonality. High variance and extreme jumps were concentrated in:

- *January, February, May, June, September*

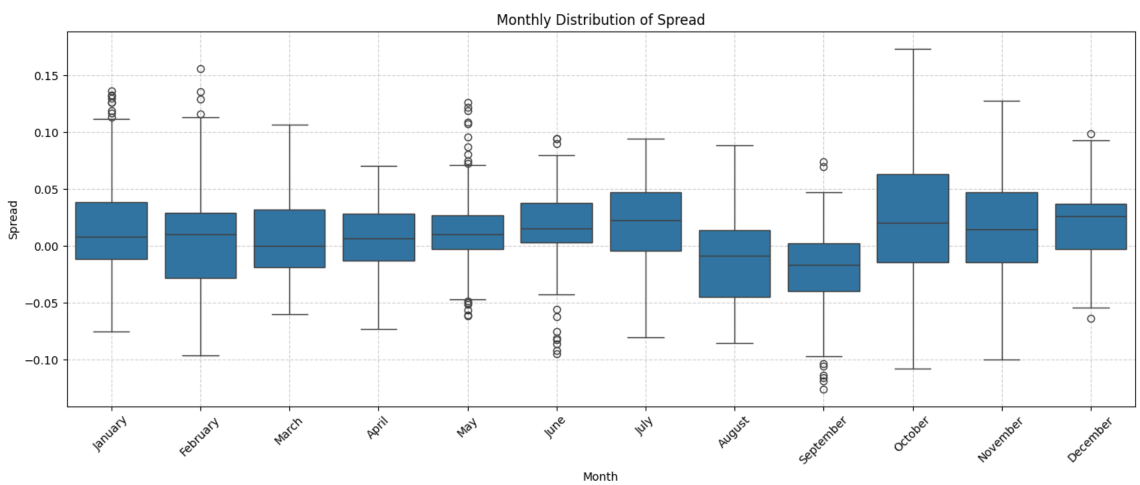

Fig 5: Monthly distribution of spread between IB USD/TND and Spot USD/TND

4.2 LSTM Neural Network

Purpose:

Forecast interbank rates in stable periods (low volatility months).

Architecture:

- **Input:** *Rolling window of last 30 IB rates*
- **Layers:** *LSTM(64) → LeakyReLU → LSTM(32) → Dense(16) → Output*
- **Optimizer:** *Adam*
- **Loss:** *MSE*
- **Regularization:** *EarlyStopping on validation loss*

Hyperparameter Tuning:

Comparative experiments were conducted on:

- *Layer width (32, 64, 128)*
- *Activation functions (ReLU, LeakyReLU, Tanh)*
- *Batch size (16, 32)*

The final config balanced performance and training stability.

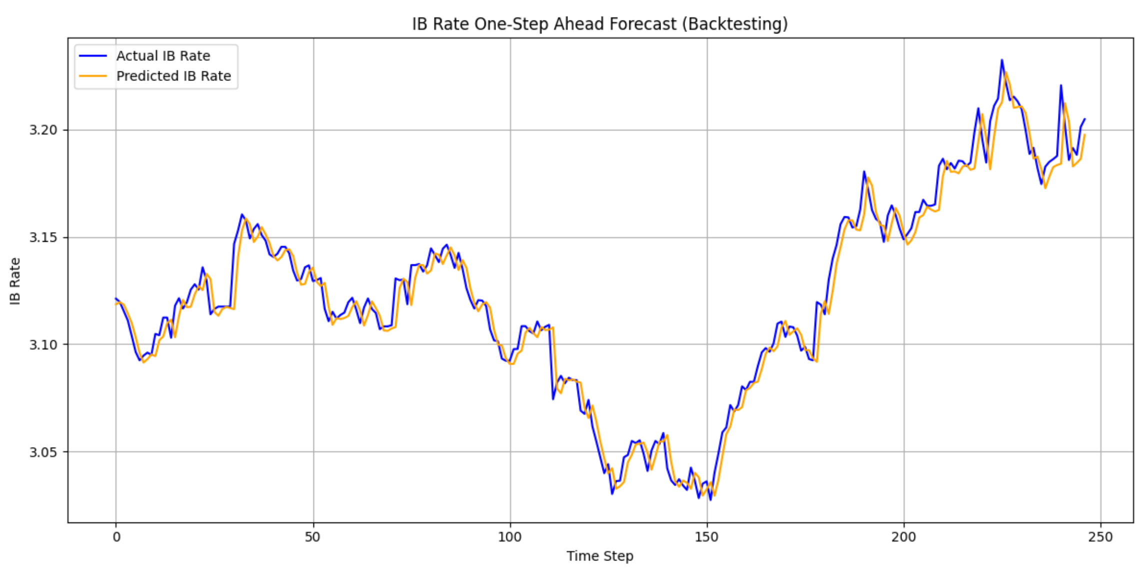

Fig 6: IB Rate One-Step Ahead Forecast using LSTM (Backtesting)

4.3 OU-Jump Diffusion Process

Purpose:

Simulate spread in volatile periods.

Model Equation: $dXt=\theta(\mu-Xt)dt+\sigma dWt+JtdNt$

Where:

- **Θ** : Speed of mean reversion
- **μ** : Long-term mean
- **σ** : Diffusion volatility
- **Jt** : Jump size (Laplace-distributed)
- **dNt** : Poisson process for jumps

Parameter Estimation:

- Estimated via **Maximum Likelihood Estimation (MLE)**
- Jump distribution: **Laplace**, chosen for its fat-tailed structure and empirical fit

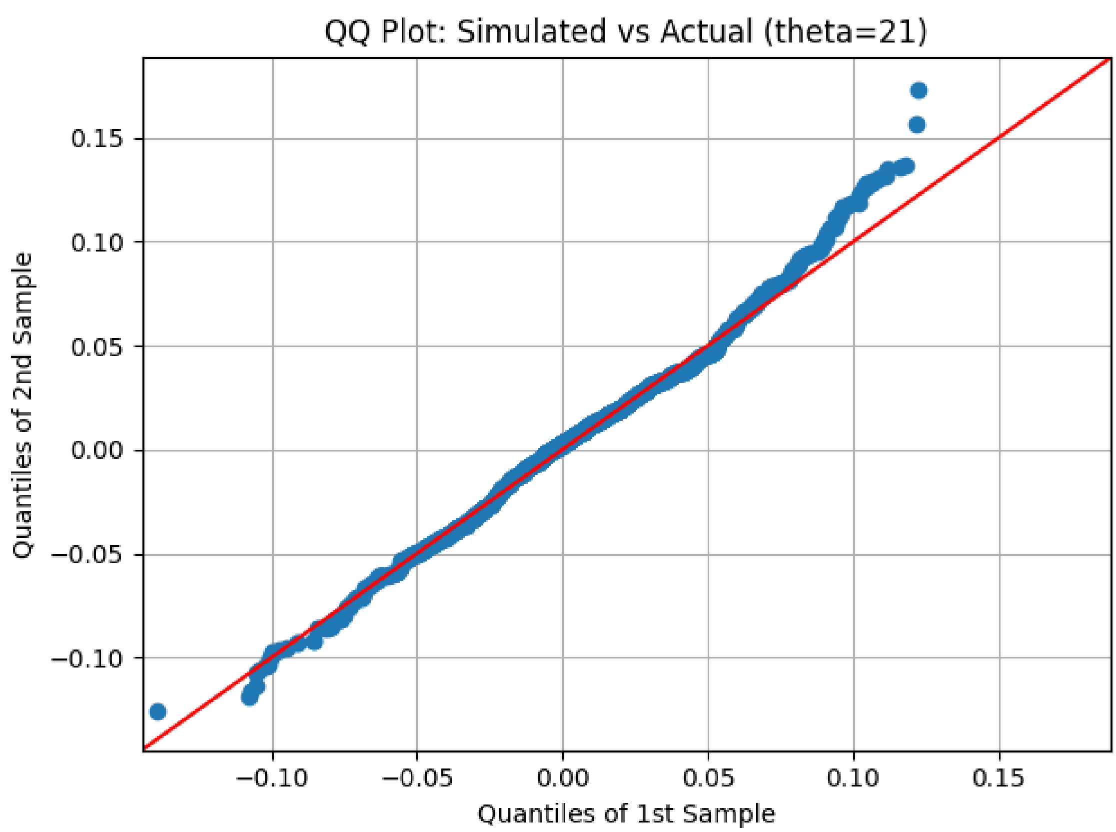

Fig 7: QQ plot of Simulated vs Actual Spread

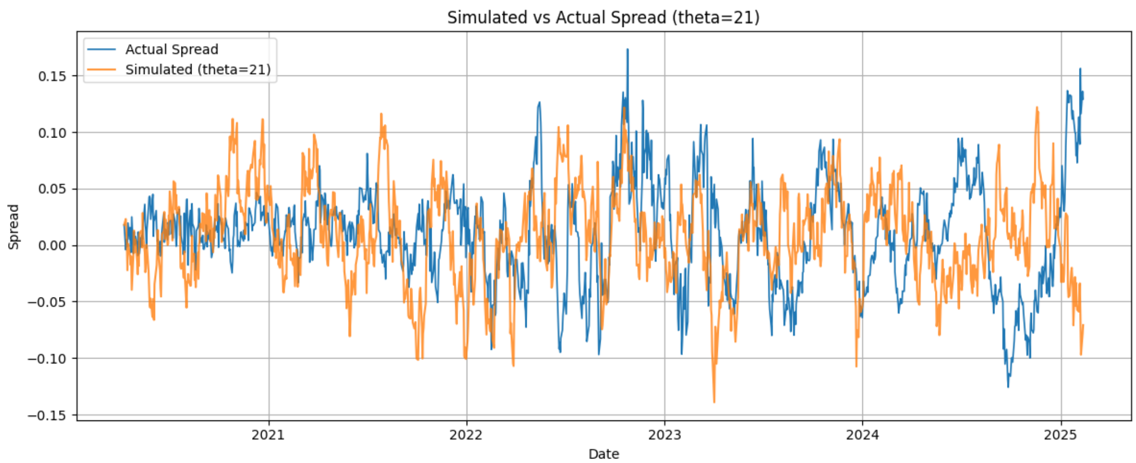

Fig 8: Simulated vs actual spread overlay

5. Hybrid Integration Logic (Integrated)

5.1 Decision Rules

For each date:

- Use **LSTM forecast** if month ∈ [3, 4, 7, 8, 10, 11, 12]
- Use **Basket + OU-Spread forecast** if month ∈ [1, 2, 5, 6, 9]
    - Only if:
        - $|\, LSTM_t - Parametric_t \,| \leq 300$ pips
        - $|\, Forecast_t - Forecast_{t-1} \,| \leq 300$ pips

Fallback to LSTM if either condition fails.

5.2 Output

- Result.xlsx contains:
    - *Forecast_LSTM*
    - *Forecast_Parametric*

- *Final_Forecast*

5.3 Model Evaluation

- **R-squared (R^2):** 0.995525
- **Mean Squared Error (MSE):** 0.00012981
- **Root Mean Squared Error (RMSE):** 0.01139321
- **Mean Absolute Error (MAE):** 0.00832613

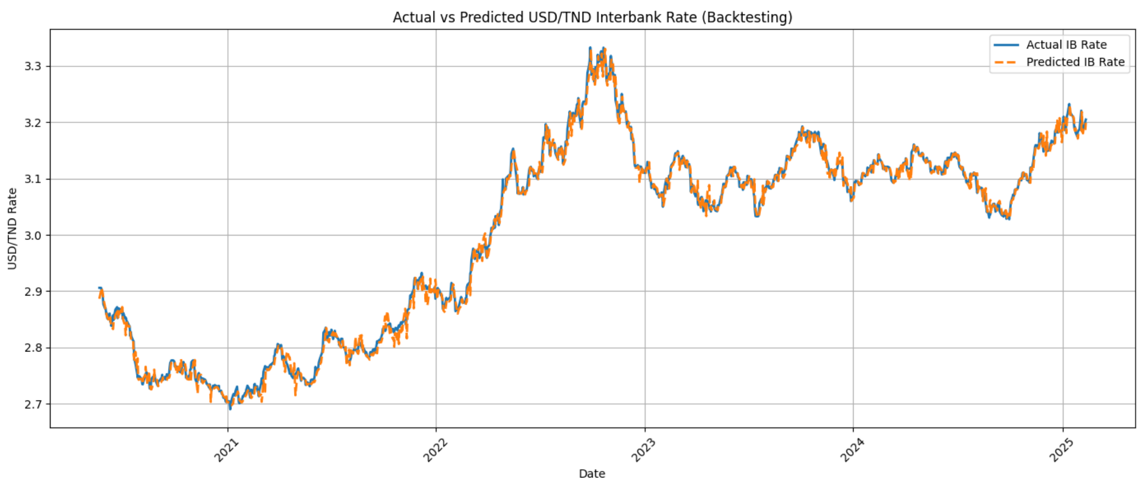

Fig 9: Actual Vs Predicted IB USD/TND rate (Backtesting)

6. Architecture Diagram

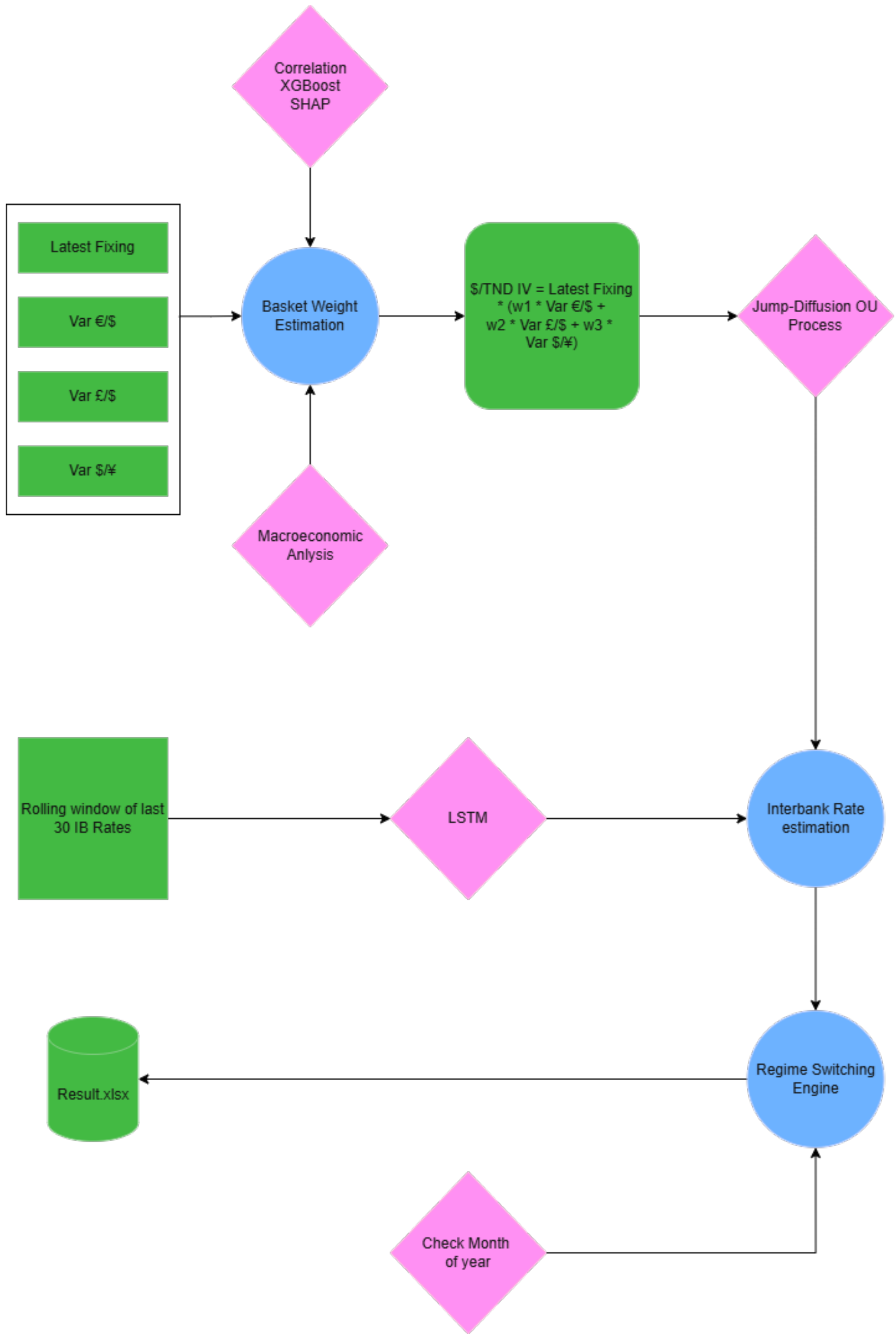

Fig 10: Real-Time IB USD/TND Forecast Model Architecture

7. Conclusion

This project successfully constructs a robust interbank rate estimator using a machine learning and stochastic modeling hybrid approach. LSTM captured short-term sequence memory. OU-Jump processes modeled liquidity distortion in volatile months. The final switching engine applied pip-thresholds to prevent unrealistic jumps.

This approach enhances transparency in FX pricing and lays a foundation for real-time market advisory tools.

Litterature Review

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